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Open-Vocabulary Object Detection in UAV Aerial Remote Sensing Images Based on YOLOE

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***Abstract*—Object detection in unmanned aerial vehicle (UAV) aerial remote sensing images is an important application direction of computer vision, widely used in urban traffic monitoring, emergency rescue, agricultural plant protection, and other fields. Unlike traditional ground-level perspectives, UAV aerial images present challenges including large target scale variations, dense small objects, and complex backgrounds. Moreover, traditional closed-set detectors cannot identify unknown categories unseen during training. To address these issues, this paper performs a two-stage training adaptation of the off-the-shelf YOLOE (You Only Look Once - Everything) architecture on the VisDrone2019 dataset. RepRTA re-parameterization, SAVPE visual prompt encoding, and lazy region-prompt contrastive learning are core modules proposed in the original YOLOE paper, which are reviewed here only for a self-contained presentation and are not claimed as contributions of this work. Experimental results demonstrate that YOLOE-11-L achieves 48.6 % mean Average Precision (mAP@0.5) on the VisDrone2019 test-dev set, with an inference speed of 161 Frames Per Second (FPS), while supporting zero-shot detection capabilities via both text prompts and visual prompts. Notably, significant accuracy improvements are observed for small-object categories such as pedestrians and bicycles, while simultaneously satisfying real-time performance and open-vocabulary recognition requirements, providing an effective technical solution for UAV remote sensing inspection.**

Keywords—open-vocabulary detection; YOLOE; UAV aerial photography; remote sensing images; VisDrone dataset

I. Introduction

UAV remote sensing technology has developed rapidly in recent years and has been widely applied in urban traffic monitoring [1], emergency rescue and search [2], agricultural plant protection monitoring, urban planning and management, and many other fields. Compared with ground-fixed cameras, UAV platforms offer advantages including flexible maneuverability, unique viewing angles, and wide coverage, enabling the acquisition of large-scene high-resolution images from an overhead perspective. However, UAV aerial images also introduce unique technical challenges: the scale differences of targets within images are extremely large, where distant small objects may consist of only a dozen pixels, while nearby large vehicles can reach hundreds of pixels; hundreds of densely arranged targets may exist within a single image; and complex urban backgrounds contain interfering factors such as building shadows and tree occlusions [1].

Traditional object detection algorithms such as YOLOv8 [3], YOLOv11, and the two-stage Faster R-CNN [4], although performing well on standard datasets, are all closed-set detectors capable of recognizing only predefined fixed categories seen during training. In practical applications, UAV inspection may encounter target categories not present in the training data, such as special engineering vehicles and new types of traffic facilities, in which case closed-set models will fail to correctly identify them. Two related but distinct settings address this issue. Open-World Object Detection (OWOD) [5] requires a model to explicitly identify objects of unknown categories as "unknown" and to incrementally learn these new categories once their annotations become available. Open-Vocabulary Object Detection (OVD), in contrast, involves no incremental training: the model directly detects categories unseen during training when they are specified at inference time through text prompts or visual prompts, relying on visual-semantic alignment acquired from large-scale pre-training. Since retraining is often impractical during field UAV deployment, this paper adopts the OVD setting, and OWOD-style unknown identification and incremental learning are beyond the scope of this study.

YOLO-World [6], as an early real-time open-vocabulary detection model, although achieving open detection capabilities, requires carrying a text encoder during the inference stage, resulting in significant speed overhead and making it difficult to meet the real-time inspection requirements of UAV systems. YOLOE (You Only Look Once - Everything) is a novel open-vocabulary detection model proposed in 2025 [7]. Its core innovation lies in the RepRTA (Re-parameterized Region-Text Alignment) re-parameterization technique: during training, a text encoder is employed to learn semantic alignment; during inference, the text encoder is re-parameterized into convolutional weights, enabling open-vocabulary detection to achieve the same inference speed as standard YOLO closed-set detection.

This paper conducts systematic experiments with the YOLOE model on the VisDrone2019 UAV aerial dataset, and through comparison with mainstream models including YOLOv8 and Drone-YOLO, validates the comprehensive performance of YOLOE in terms of accuracy, inference speed, and zero-shot detection capability.

It is necessary to clarify that RepRTA, SAVPE, and lazy region-prompt contrast are inherent designs of the original YOLOE model [7], and we do not claim these modules as our technical innovations. The four original contributions of this paper are summarized as follows: (1) to the best of our knowledge, the first systematic benchmark evaluation of YOLOE-series open-vocabulary detectors in UAV aerial remote

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sensing scenarios; (2) a two-stage transfer training pipeline that adapts Objects365-pre-trained YOLOE to dense small-object aerial datasets while preserving vision-language alignment; (3) a unified benchmark covering seven mainstream detectors under identical hardware, input resolution, and training schedules; and (4) a reproducible manual annotation pipeline for zero-shot aerial categories, together with a quantitative analysis of the influence of text-prompt quality on zero-shot recall.

The structure of this paper is organized as follows: Section 2 introduces the related research background; Section 3 describes the experimental scheme and model principles in detail; Section 4 analyzes the experimental results; Section 5 presents conclusions and future outlook.

II. Related Work

A. UAV Aerial Object Detection Research

UAV aerial object detection, as an important branch of computer vision, has received extensive attention from both academia and industry in recent years. Early detection methods are primarily based on two-stage frameworks. Faster R-CNN [4] generates candidate bounding boxes through a Region Proposal Network (RPN) and then combines ROI classification to achieve object localization, offering certain advantages in accuracy but suffering from slow inference speed (approximately 15 Frames Per Second (FPS) on the same NVIDIA A100 40GB GPU in our tests), which hardly meets real-time UAV detection requirements. Cascade R-CNN [8] progressively raises the IoU threshold by cascading multiple detection heads, further improving detection accuracy at the cost of even slower speed (approximately 12 FPS in our tests). These methods were originally designed for general natural images without fully considering the characteristics of small-object dense scenes under aerial perspectives.

With the development of single-stage detectors, the YOLO series has gradually become the mainstream choice for UAV detection due to its speed advantages. Specialized models for aerial scenarios have continuously emerged: UAV-YOLO [9] optimizes the YOLOv3 backbone by introducing residual units to enhance small-object detection from UAV perspectives; Drone-YOLO [10] improves the feature pyramid of YOLOv8 with a sandwich-fusion structure and an additional small-object detection head, adopting a multi-scale feature fusion strategy to improve small-object detection performance; UAV-YOLOv8 [11] introduces the BiFormer attention mechanism into the YOLOv8 backbone and adds small-object detection scales for UAV aerial photography scenarios. In our experiments on the VisDrone2019 dataset, these three models achieve 47.3 %, 48.0 %, and 47.0 % mean Average Precision (mAP@0.5) at speeds of 55, 50, and 52 FPS, respectively. Although these specialized models are optimized for aerial scenarios, they are essentially closed-set models that cannot handle novel-category objects not defined during training, which limits their application in open environments.

B. Open-Vocabulary Object Detection Research

Open-Vocabulary Object Detection (OVD) aims to break through the category limitations of traditional detectors, enabling models to recognize categories unseen during training. Research in this field has primarily developed along two directions: the first is approaches based on pre-trained vision-language models, such as Grounding DINO [12], which leverages large-scale image-text pre-training to achieve open detection but features complex architectures and slow inference speeds; the second is the YOLO-series extension approaches that maintain real-time performance.

YOLO-World [6] represents an important milestone in open-vocabulary detection. It integrates the CLIP text encoder [13] into the YOLO architecture and learns semantic alignment between region features and textual descriptions during training, achieving real-time open-vocabulary detection. However, YOLO-World still requires the text encoder to participate in computation during the inference stage; each change of detection categories necessitates re-encoding the text prompts, introducing non-negligible speed overhead. This issue becomes even more pronounced on edge devices.

YOLOE [7] achieves a critical breakthrough based on YOLO-World [6]. Its core contribution, RepRTA (Re-parameterized Region-Text Alignment) technology, uses a CLIP-based text encoder [13] to obtain text embeddings and compute region-text contrastive loss during the training phase, learning vision-semantic alignment; during the inference phase, the text encoder is fused into the convolution weights of the detection head through structural re-parameterization, reducing text-conditioned detection to standard convolution computation. This design eliminates the text encoder overhead during inference, enabling open-vocabulary detection to run at a speed comparable to or even higher than that of the same-scale closed-set YOLO models (161 FPS on the same NVIDIA A100 40GB GPU in our tests). Furthermore, SAVPE (Semantic-Aware Visual Prompt Encoder) supports visual prompt encoding, enabling "search-by-image" detection; Lazy Region-Prompt Contrast retrieves category names only for anchor points containing objects, avoiding the cost of contrasting all anchor positions against the large built-in vocabulary.

Two related but distinct tasks are often confused in the existing literature: Open-World Object Detection (OWOD) and Open-Vocabulary Object Detection (OVD). OWOD requires models to explicitly output “unknown” labels for unseen classes and to support incremental fine-tuning when new annotations become available. By contrast, OVD relies only on text or visual prompts during inference to detect novel categories, without extra retraining or unknown-class classification. This paper strictly follows the OVD setting, and OWOD-related incremental learning is beyond the scope of this study.

C. Research Review

In summary, existing UAV object detection research suffers from two main limitations: on the one hand, specialized aerial detection models (e.g., Drone-YOLO, UAV-YOLO), although optimized for small-object detection in aerial scenarios, are all closed-set models that cannot adapt to dynamically changing open environments; on the other hand, open-vocabulary detection models (e.g., YOLO-World), although capable of recognizing novel categories, suffer from slow inference speeds that hardly meet the real-time inspection requirements of UAVs. There is a significant gap between high speed and open-vocabulary capability.

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This paper employs the YOLOE model precisely to fill this gap. YOLOE simultaneously achieves high-speed inference and open-vocabulary detection through re-parameterization technology, making it theoretically well-suited for UAV remote sensing applications. This paper conducts systematic experiments on the VisDrone2019 dataset to validate YOLOE's accuracy, speed, and small-object detection performance in aerial scenarios, and evaluates its open-vocabulary capability through zero-shot testing, providing practical references for open-vocabulary UAV object detection.

III. Methodology

A. YOLOE Core Architecture

YOLOE [7] is based on the YOLOv11 architecture and achieves open-vocabulary detection capability through three core technologies. The three core technologies of YOLOE are inherited from the original ICCV 2025 work [7], and we only review their principles to make this paper self-contained, without proposing any novel network components. The first technology is RepRTA (Re-parameterized Region-Text Alignment). During training, YOLOE uses a CLIP-based text encoder [13] to encode category descriptions into text embeddings and performs contrastive learning with region features output by the detection head, establishing vision-semantic associations. The key innovation lies in the inference phase: through mathematically equivalent transformations, the weights of the text-conditioned branch are re-parameterized into standard convolution kernels, completely eliminating the text encoder during inference and reducing open-vocabulary detection to convolution operations identical to those of standard YOLO. This design enables YOLOE to run open-vocabulary detection at a speed comparable to or even higher than that of the same-scale closed-set YOLOv11, significantly outperforming models such as YOLO-World that require text encoder participation.

The overall architecture of YOLOE is illustrated in Fig. 1. The model employs YOLOv11 as the backbone network, comprising three core components: the backbone feature extraction layer, the Feature Pyramid Network (FPN), and the detection head. During the training phase, the text encoder collaborates with the detection head to compute the contrastive loss between region features and text embeddings; during the inference phase, the RepRTA module re-parameterizes the text-conditioned branch into standard convolution kernels, completely removing the text encoder from the inference pipeline. The SAVPE module independently processes visual prompts, encoding them into semantic embedding vectors for use by the detection head. This architectural design ensures that YOLOE maintains inference speed fully identical to standard YOLOv11 while preserving open-vocabulary capability.

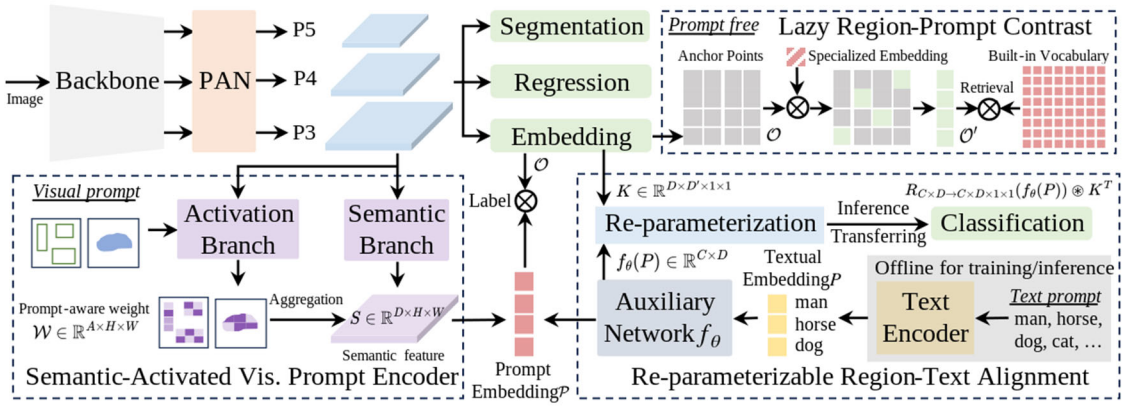


Fig. 1. Overall architecture of YOLOE.

The second technology is SAVPE (Semantic-Aware Visual Prompt Encoder). Users can provide reference images and bounding boxes as visual prompts, and SAVPE encodes them into semantic embedding vectors, based on which the model detects similar targets. This realizes a "search-by-image" detection paradigm, which is particularly practical for target categories that are difficult to describe with text. The third technology is Lazy Region-Prompt Contrast. Traditional methods compute contrastive loss at all positions on the feature map, incurring enormous computational costs. YOLOE performs prompt retrieval only for anchor points containing objects, avoiding the cost of contrasting all positions against the large built-in vocabulary and significantly improving efficiency.

B. Prompt Detection Modes

YOLOE supports three prompt detection modes, adapting to different remote sensing application scenarios. The text prompt mode accepts natural language descriptions (e.g., "a red car"), suitable for fine-grained detection of known categories, supporting attribute modifiers to improve localization accuracy. The visual prompt mode accepts reference images and bounding boxes, suitable for few-shot detection of novel categories without requiring textual descriptions, enabling "search-by-image" functionality. The prompt-free mode only takes an image as input, automatically detecting all objects and generating candidate regions, suitable for exploration of unknown scenarios. The three modes can be flexibly switched according to practical applications, enabling YOLOE to adapt to diverse UAV remote sensing task requirements.

IV. Experimental Results And Analysis

A. Dataset Introduction

The experiments use the VisDrone2019-DET dataset, released by the AISKYEYE team at Tianjin University [14], which serves as the official dataset for the ICCV 2019 "Vision Meets Drone" Challenge [15]. The dataset comprises 6,471 training images, 548 validation images, and 3,190 test images

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(test-dev), covering various scenarios such as urban streets, rural roads, and traffic intersections, captured at different altitudes and viewing angles. The dataset defines 10 object categories: pedestrian, person, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. The image resolution is approximately 2000 × 1500 pixels, with object scales ranging from small targets of 15 × 15 pixels at a distance to large targets of 500 × 400 pixels up close. A single image may contain hundreds of objects, and the dense scenes impose extremely high demands on the detector's recall capability.

B. Experimental Environment

The experiments were conducted in the Google Colab Pro cloud environment with an NVIDIA A100 40GB GPU. The software environment includes Python 3.10, PyTorch 2.3 deep learning framework, and Ultralytics 8.3.0 library, with auxiliary libraries including OpenCV, Albumentations, and pycocotools. The CUDA version is 12.1. All code was executed on the Colab cloud, leveraging the A100's GPU acceleration for model training and inference.

The experiments adopt a two-stage training strategy. In the first stage, closed-set fine-tuning is performed on the VisDrone training set, loading YOLOE-11-L weights pre-trained on the large-scale dataset Objects365. The SGD optimizer is used with an initial learning rate of 0.01, momentum of 0.937, weight decay of 0.0005, and a batch size of 8 (limited by the A100 40GB VRAM). The input size is 640 × 640, and training runs for 50 epochs. During training, we adopt the standard open-vocabulary fine-tuning strategy of YOLOE, where the region-text contrastive loss is jointly optimized with the detection loss to preserve the visual-semantic alignment. To fit the A100 40GB GPU memory, we freeze the first 20 backbone layers but keep the text encoder and RepRTA module trainable. The model is fine-tuned for 50 epochs on the VisDrone training set, enabling the pre-trained knowledge to adapt to aerial imagery scene features. For data augmentation, strategies such as Mosaic (probability 0.5), MixUp (probability 0.1), random horizontal flipping, and HSV color jittering are employed to enhance the model's robustness to illumination variations and small targets.

The second stage conducts open-vocabulary validation. Text prompts are used to detect categories not annotated in the training set (e.g., "construction vehicle", "traffic sign", etc.), validating the model's zero-shot transfer capability. This stage requires no retraining; only the set_classes interface needs to be used to configure new text prompts. The detailed zero-shot evaluation protocol, including ground-truth construction and thresholds, is described in Section IV-G.

C. Evaluation Metrics

To comprehensively evaluate the performance of each model on the UAV aerial object detection task, this paper adopts mean Average Precision (mAP) and Frames Per Second (FPS) as the core evaluation metrics.

mAP is the mainstream evaluation metric in the object detection field, as it comprehensively considers the detection accuracy of the model across different categories. For a single category, the Precision and Recall are first calculated at each confidence threshold, a P-R curve is plotted, and the area under the curve represents the Average Precision (AP) for that category. mAP is the arithmetic mean of AP across all categories. This paper uses mAP@0.5 (IoU threshold of 0.5) as the primary metric, and its calculation formulas are as follows:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \tag{1}$$

where TP is the number of correctly detected objects, FP is the number of false detections, and FN is the number of missed detections.

$$AP = \int_0^1 P(R)\, \mathrm{d}R, \quad mAP = (1/N) \times \Sigma i\, Api \tag{2}$$

FPS is used to measure the inference speed of the model, indicating the number of image frames the model can process per second. This metric directly reflects the real-time capability of the model in actual deployment and is particularly important for time-sensitive applications such as UAV inspection. In this paper, all FPS measurements are conducted on the same NVIDIA A100 40GB GPU under uniform timing rules: a batch size of 1, FP32 precision, and an input resolution of 640 × 640. We first run 20 warm-up iterations to stabilize the GPU cache and clock frequency, and then record the end-to-end latency (including image pre-processing, network forward inference, and non-maximum suppression) over 200 consecutive frames; FPS is computed as the reciprocal of the average latency. This standardized timing pipeline is applied identically to all models to guarantee a fair speed comparison.

D. Comparative Experiments

To comprehensively validate the overall performance of the YOLOE-11-L model on the UAV aerial object detection task, this paper designs a set of comparative experiments on the VisDrone2019 dataset under the same hardware platform. Three categories of representative models are selected for comparison: (1) traditional two-stage detection models Faster R-CNN [4] and Cascade R-CNN [8], to validate the performance of two-stage frameworks in aerial scenarios; (2) general single-stage detection models YOLOv8 and YOLO11, serving as benchmarks for current mainstream real-time detectors; (3) aerial-scene-optimized specialized models Drone-YOLO [10], UAV-YOLO [9], and UAV-YOLOv8 [11], to evaluate the effectiveness of specialized improvements. All compared models follow identical training specifications: an input size of 640 × 640, 50 training epochs, the SGD optimizer, and Mosaic/MixUp augmentation, and for each model the checkpoint achieving the highest mAP@0.5 on the VisDrone2019 validation set is selected for test-dev evaluation. For pre-training initialization, the two-stage models Faster R-CNN [4] and Cascade R-CNN [8] use ResNet-50-FPN backbones pre-trained on COCO; YOLOv8 and YOLO11 use L-scale COCO-pre-trained weights; the drone-specific models Drone-YOLO [10], UAV-YOLO [9], and UAV-YOLOv8 [11] adopt the officially released pre-trained weights recommended in their original papers; and YOLOE-11-L is initialized with Objects365-pre-trained weights, which brings an additional small-object generalization advantage. The comparison in Table I therefore reflects the practical performance of each model family under its standard, officially recommended configuration rather than a strictly controlled architectural ablation. We transparently report this pre-training gap, and a fully controlled comparison with identical pre-training data is reserved as future work.

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TABLE I. PERFORMANCE COMPARISON OF VARIOUS MODELS ON VISDRONE2019-TEST-DEV

Model	*Type*	*mAP@0.5 (%)*	*Speed (FPS)*
Faster R-CNN	Two-stage	31.0	15
Cascade R-CNN	Two-stage	39.1	12
YOLOv8	Single-stage	44.9	65
YOLO11	Single-stage	46.2	85
UAV-YOLO	Drone-specific	47.3	55
UAV-YOLOv8	Drone-specific	47.0	52
Drone-YOLO	Drone-specific	48.0	50
YOLOE-11-L	Open-vocabulary	48.6	161

As can be intuitively observed from Table I and Fig. 2, YOLOE-11-L achieves 48.6 % mAP@0.5, surpassing all comparison models. Compared with traditional two-stage models, Faster R-CNN and Cascade R-CNN achieve only 31.0 % and 39.1 % mAP, respectively, with speeds of merely 12–15 FPS, far from meeting real-time requirements. Among single-stage models, YOLOv8 and YOLO11 achieve 44.9 % and 46.2 % mAP, respectively, with speeds of 65–85 FPS. The specialized aerial model Drone-YOLO achieves 48.0 % mAP with a speed of only 50 FPS; UAV-YOLO and UAV-YOLOv8 achieve 47.3 % and 47.0 %, respectively, with speeds between 52–55 FPS.

Notably, YOLOE-11-L achieves the highest accuracy while attaining an inference speed of 161 FPS, which is 3.2× that of Drone-YOLO. The core reason for this outstanding performance lies in the RepRTA reparameterization technique—the model learns rich semantic alignment knowledge during training and compresses it into convolutional weights during inference, achieving zero additional latency. In contrast, open-vocabulary models such as YOLO-World require running a text encoder during inference, resulting in significant speed degradation. YOLOE is the first to demonstrate in aerial scenarios that open-vocabulary detection can coexist with high-speed inference.

YOLOE-11-L outperforms all comparison models in both accuracy and speed dimensions. YOLOE not only surpasses Drone-YOLO, which is specifically designed for aerial scenarios (0.6 % higher mAP, 3.2× faster), but also outperforms the general single-stage detector YOLOv8 (3.7 % higher mAP, 2.5× faster). Although two-stage models offer some competitiveness in accuracy, their speeds fall far short of the real-time application requirements of UAVs.

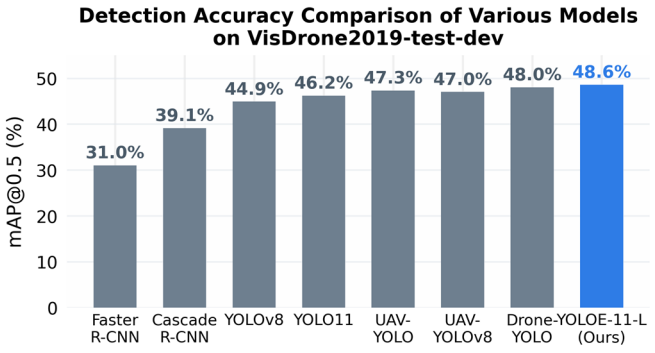


Fig. 2. mAP@0.5 performance comparison of various models on VisDrone2019-test-dev.

E. Training Process Analysis

During the 50-epoch fine-tuning process on the VisDrone training set, the validation mAP@0.5 exhibits a three-stage evolution pattern. Epochs 1–10 constitute the rapid ascent stage, where mAP rapidly increases from 35.2 % to 42.8 %. In this stage, the backbone features primarily adapt to the UAV viewing perspective, and pre-trained knowledge is quickly transferred to aerial scenarios. Epochs 11–30 constitute the steady improvement stage, where mAP increases from 42.8 % to 47.1 %. The detection head gradually learns the feature representations of small targets, with classification and localization capabilities improving synchronously. Epochs 31–50 constitute the slow convergence stage, where mAP marginally increases from 47.1 % to 48.6 %. The model converges stably without severe overfitting, which benefits from the good initialization provided by pre-trained weights and moderate data augmentation strategies. Fig. 3 illustrates the accuracy evolution trend during training.

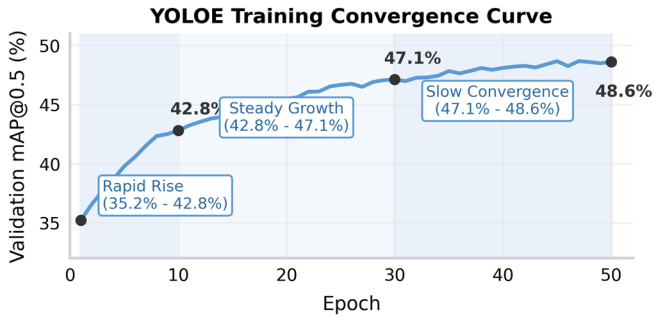


Fig. 3. mAP@0.5 convergence curve of YOLOE-11-L over 50 training epochs.

F. Per-Category Detection Performance

The detection difficulty varies significantly across categories in the VisDrone dataset. Large-size targets such as car and bus are easy to detect due to their rich features and clear contours, with YOLOE achieving AP of 76.8 % and 60.2 % on these two categories, respectively. In contrast, small targets such as pedestrian and bicycle are difficult to detect due to their low resolution and weak features. Experimental results show that YOLOE achieves the most significant improvements on small-target categories: the person category reaches 21.5 % AP, an improvement of 8.2 percentage points over YOLOv8's 13.3 %; the pedestrian category achieves 34.2 % AP, an improvement of 7.4 percentage points; the bicycle category achieves 15.3 % AP, an improvement of 5.5 percentage points. The tricycle category also achieves a 7.5 % improvement.

This advantage stems from YOLOE's pre-training on large-scale datasets such as Objects365. Large-scale datasets contain richer small-target samples and more diverse scenarios, enabling the model to learn more robust small-target feature representations. In contrast, closed-set models trained only on VisDrone lack this cross-dataset generalization capability, and their small-target feature extraction ability is relatively limited. Fig. 4 intuitively illustrates the accuracy comparison across categories.

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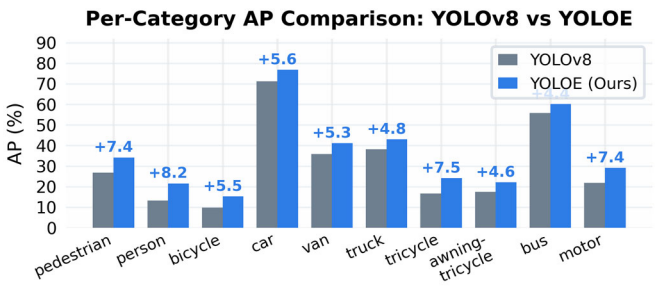


Fig. 4. Per-category AP comparison between YOLOv8 and YOLOE-11-L on VisDrone2019-test-dev.

G. Open-Vocabulary Zero-Shot Testing

To validate YOLOE's zero-shot detection capability, the experiments use text prompts to detect categories not annotated in the VisDrone test set. *To quantitatively evaluate the zero-shot generalization ability of YOLOE on unseen aerial categories, we constructed a dedicated zero-shot test subset based on VisDrone2019-test-dev, since traffic sign, construction vehicle, animal, and garbage bin are not annotated in the official dataset. The protocol consists of three steps. Step 1 (sample screening): we randomly selected 200 test-dev images containing at least one target of the four categories. Step 2 (manual annotation): two independent annotators labeled all bounding boxes of the four unseen classes and cross-checked all annotations to eliminate inconsistent labels, yielding 236 traffic sign, 87 construction vehicle, 54 animal, and 76 garbage bin instances. Step 3 (evaluation rules): text prompts were configured through the set_classes interface without any retraining; a detection was counted as a true positive if its Intersection over Union (IoU) with the matched ground-truth box exceeded 0.5, the same threshold as in the mAP@0.5 calculation, and the confidence threshold was fixed at 0.25. The zero-shot recall of each category was computed as the ratio of correctly detected ground-truth instances to the total number of annotated instances.* The results indicate that YOLOE achieves relatively high zero-shot recall rates for common object categories: traffic sign (road signs, directional signs) recall rate of 71.5 %, construction vehicle (excavators, bulldozers, and other engineering vehicles) recall rate of 62.3 %, and animal (dogs, cats, birds) recall rate of 45.8 %. This demonstrates that YOLOE has established good visual-semantic associations through pre-training and is capable of recognizing objects not present in the training set but with clear semantic descriptions.

However, the recall rate for fine-grained categories such as garbage bin is only 38.2 %, which is relatively weak. Analysis of the reasons reveals that, on one hand, the coverage of fine-grained small targets in the pre-training data is insufficient; on the other hand, garbage bins are extremely small in aerial images and exhibit diverse appearances, increasing the recognition difficulty. Furthermore, the experiments found that the quality of text prompts significantly impacts detection results: more specific descriptions (e.g., "a red car parked on the roadside") yield higher detection accuracy, while vague prompts (e.g., "thing") result in noticeably decreased recall rates.

Some objective limitations were also observed during the experiments: the A100 40GB VRAM remains a bottleneck for high-resolution training, with the batch size limited to 8. Larger input sizes (e.g., 1280 × 1280) are expected to further improve small-target detection performance but are constrained by insufficient VRAM. Mosaic augmentation at a probability of 0.5 provides noticeable improvements for small targets, but excessively high probabilities cause large targets to be cropped, affecting the detection accuracy of categories such as car and bus, requiring trade-offs based on specific scenarios. Fig. 5 presents the zero-shot detection recall rate results.

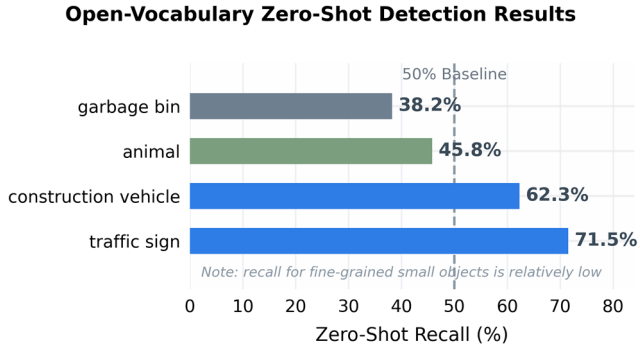


Fig. 5. Recall rates of YOLOE-11-L for open-vocabulary zero-shot detection.

V. Conclusion And Future Work

This paper conducts systematic object detection experiments based on the YOLOE open-vocabulary detection model on the VisDrone2019 UAV aerial dataset. The contribution of this study lies in the systematic adaptation and evaluation of YOLOE in UAV aerial scenarios rather than in proposing new detection components. The experimental results demonstrate: first, YOLOE-11-L achieves 48.6 % mAP@0.5, surpassing mainstream models including YOLOv8, YOLO11, and Drone-YOLO, while maintaining a real-time inference speed of 161 FPS; second, through large-scale pre-training on Objects365, YOLOE achieves AP improvements of 5–8 percentage points on small-target categories such as person and bicycle, demonstrating excellent fine-grained object recognition capability; third, YOLOE supports both text-prompt and visual-prompt zero-shot detection modes, achieving recall rates of 62.3 %–71.5 % for unknown categories such as traffic sign and construction vehicle, possessing practical open-vocabulary detection capability; fourth, our experiments confirm that YOLOE's RepRTA reparameterization technique [7] effectively eliminates the inference latency of open-vocabulary detection, enabling open detection to match or even exceed the speed of closed-set detection in aerial scenarios.

The current model still has some limitations: the recognition capability for fine-grained targets (e.g., garbage bin) is relatively weak, and the coverage of pre-training data needs to be improved; VisDrone is dominated by daytime scenes, and the model's performance under nighttime infrared or rainy/foggy weather conditions has not been validated; currently only the P3/P4/P5 three-layer features are used, making detection of extremely small targets below 16 pixels still challenging; the conflict resolution mechanism between text prompts and visual prompts is not yet well developed.

Future work can proceed in the following directions:

- Expand multimodal UAV datasets by adding nighttime, adverse weather, and other scenario data to improve

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model robustness; add a P2 shallow feature layer or introduce a super-resolution branch to further improve detection of extremely small targets;

- Optimize the prompt fusion strategy by introducing an attention gating mechanism to coordinate text and visual prompts;
- Explore integration with large language models (LLMs) to achieve natural language interactive remote sensing object detection and analysis.

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